Jonas KOEPPEL

Postdoctoral scholar in the Shendure and Pinglay labs

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Education

10.2019 – 02.2024 PhD student in genomics at the Wellcome Sanger Institute and Cambridge University

10.2016 – 08.2019 Master of Science in Molecular Bioscience, Major Cancer Biology at the German Cancer Research Center (DKFZ) and Heidelberg University (Grade 1.0, best possible)

10.2013 – 09.2016 Bachelor of Science in Biosciences at Heidelberg University (Grade 1.2, “very good”)

09.2012 – 06.2013 Study of Traditional Chinese Painting & Calligraphy at Shanghai Normal University, PR China

Selected research projects

09.2024 – present **Postdoctoral fellow: Saturation deletion scanning of the human genome**

Shendure and Pinglay labs – University of Washington, Seattle, US

- This work aims to test the functional consequences of all sequences in the human genome by creating highly multiplexed deletions in human cell lines

02.2024 – 07.2024 **Research associate: Scaling genome scrambling methods**

Parts lab – Wellcome Sanger Institute, Cambridge, UK

06.2020 – 02.2024 **PhD project: Randomizing the human genome by engineering recombination between repeat elements**

Parts lab – Wellcome Sanger Institute, Cambridge, UK

- This work generated a rich toolbox to manipulate genome architecture at the megabase scale.
- Targeted high copy-number elements with CRISPR prime editing to precisely insert hundreds of recombinase sites into a single human genome.
- Induced thousands of structural variants in the and mapped the selection pressures acting on them

06.2020 – 02.2024 **PhD project: Randomizing gene regulatory regions using prime editing**

Parts lab – Wellcome Sanger Institute, Cambridge, UK

- Engineered efficient prime editing cell lines and tiled regulatory regions with recombinase sites.
- Established targeted nanopore sequencing methods to read out structural variation.
- Discovered a novel enhancer cluster controlling the OTX2 gene and experimentally assessed the impact of shuffling and deleting enhancers within the cluster.

06.2020 – 02.2023 **PhD project: Prediction of prime editing insertion efficiencies using sequence features and DNA repair determinants**

Parts lab – Wellcome Sanger Institute, Cambridge, UK

- Characterized the efficiencies for thousands of small insertions with prime editing, uncovered the determinants governing the rates and built a computational model to predict efficiencies.
- Discovered a novel repair pathway involved in prime editing.
- Resulted in a first author publication in *Nature Biotechnology* [LINK].

01.2018 – 11.2020 **Small-molecule induced polymerization triggers the degradation of BCL6**

Seattle, USA March, 2025

Ebert lab – Broad Institute of Harvard and MIT / Dana Farber Cancer Institute, Harvard Medical School, MA, USA & Fröhling lab – National Center for Tumor Diseases / German Cancer Research Center, Heidelberg, BW, Germany

- Elucidated the mechanism of action of a small-molecule using CRISPR screens and biochemistry.
- Discovered targeted protein polymerization followed by degradation as a novel drug modality.
- Resulted in a US patent application and a shared first author publication in *Nature* [LINK].

Extracurricular projects

02.2022 – 02.2024 **Entrepreneurship**

- Coordinated the filing of a patent between the technology transfer office, our lab and collaborators in the Church lab at Harvard Medical School on the genome randomization method developed in my PhD.
- Established connections with partners from Hummingbird, Civilization Ventures, and Fifty Years for potential future commercialization routes.

10.2016 – 05.2018 **Mentor for the 'Jugend Forscht' high school science fair (2018)**

- Our team finished national 3rd place out of over 1200 participants in the field of Biology.
- We constructed, expressed and validated light-gated proton channels in *E. coli* to rapidly and reversibly decouple the respiratory chain from ATP production and control bacterial growth.

10.2017 – 06.2018 **Organized the weekly 'Data Analysis and Programming with R' course**

- Initiated a self-study R course to complement the limited offerings of our master's at the time.
- Contacted Data Camp who provided us with free licenses for courses on their platform.
- Gave a three-session workshop on the analysis of large screen and mutation data sets.

Teaching experience

02.2022 – 12.2022 **Mentorship/Supervision**

- Mélanie Gouley (Master's thesis – European School of Biotechnology, Strasbourg). The thesis project was graded with the best possible mark by independent examiners and Mélanie secured a Ph.D. position at ETH Zürich.
- Valentin Reberning (Bachelor's thesis – University in Wiener Neustadt). Defense anticipated in July

10.2020 – 07.2021 **Supervising Cambridge University students in "Cell and Developmental Biology"**

Weekly supervisions in small groups (2 and 3 students). Discussing course material, assigning and grading essays, preparing mock-exams, introducing students to relevant research.

04.2015 – 07.2016 **Teaching assistant for the practical laboratory course "Methods of Molecular Bioscience"**

- Demonstrated the experiments (genomic and plasmid DNA preparation, agarose gel electrophoresis, SDS-PAGE) and presented the theory in introductory lectures before each session.
- Graded reports and prepared exam preparation tutorials.

Stipends

05.2023	Pembroke College Grant (£1,156 covering expenses for the Biology of Genomes conference)
04.2022	Keystone symposia scholarship ($1200 covering conference expenses)
10.2019 – present	Wellcome Sanger Institute stipend (£22,189/yr covering my PhD at the Wellcome Sanger Institute)
01.2018 – 05.2018	PROMOS stipend (€2,575. German Academic Exchange Service (DAAD))
10.2016 – 10.2018	Germany stipend (€7,200. Federal Government of Germany and Prof. Helmut Determann)

Skills

Seattle, USA March, 2025

Experimental	**Genome engineering:** CRISPR Prime editing and knockout. CRISPR-associated transposases. Functional genomics: Arrayed and pooled screens with proliferation or sorting-based readouts. Sequencing: Short (Illumina) - and long-read (ONT) whole genome and targeted sequencing.
Computational	**Computational:** R, Bash, compute clusters (LSF). Analysis workflows: Whole-genome sequencing, ATAC and RNA sequencing, CRISPR screens. Starting from fastq files, alignment (bwa-mem, minimap2, NGMLR), SNV and SV calling (caveman, pindel, gridss, brass, sniffles2, nanomonsv, savana), assembly (flye), variant classification (R). RNA seq (salmon)
Languages	German (native), Chinese (spoken: mother tongue, written HSK 4), English (IELTS 8.5/9)

Scientific contributions

First and shared first author publications

J Koeppel*, R Ferreira*, T Vanderstichele, LM Riedmayr, EM Peets, G Girling, J Weller, FG Liberante, T Ellis, GM Church, L Parts. (2025) "Randomizing the human genome by engineering recombination between repeat elements". **Science**.

J Koeppel*, P Murat*, G Girling, EM Peets, M Gouley, V Reberning, A. Maheshwari, J Hepkema, J Weller, JHJ Jebaraj, R Crawford, FG Liberante, L Parts. (2024) "Resolution of a human super-enhancer by targeted genome randomisation". **bioRxiv**.

J Koeppel*, J Weller*, EM Peets*, A Pallaseni, I Kuzmin, U Raudvere, H Peterson, F Liberante, L Parts (2023). "Prediction of prime editing insertion efficiencies using sequence features and DNA repair determinants". **Nature Biotechnology**.

M Słabicki*, H Yoon*, **J Koeppel***, L Nitsch, SR Burman, CAD Genua, KA Donovan, AS Sperling, M Hunkeler, JM Tsai, R Sharma, A Guirguis, C Zou, P Chudasama, JA Glasser, PG Miller, C Scholl, S Fröhling, RP Nowak, ES Fischer, BL Ebert (2020). "Small molecule-induced polymerization triggers degradation of BCL6". **Nature**.

Co-author publications

S Pinglay, JB Lalanne, RM Daza, **J Koeppel**, XY Li, DS Lee and J Shendure (2025). "Multiplex generation and single cell analysis of structural variants in a mammalian genome". **Science**.

M Coelho, M Strauss, A Watterson, S Cooper, S Bhosle, G Illuzzi, E Karakoc, C Dinçer, S F Vieira, M Sharma, M Moullet, D Conticelli, **J Koeppel**, […], M Garnett (2024). "Base editing screens define the genetic landscape of cancer drug resistance mechanisms". **Nature Genetics**.

J Weller*, A Pallaseni*, **J Koeppel**, L Parts (2023). "Predicting Mutations Generated by Cas9, Base Editing, and Prime Editing in Mammalian Cells". **The CRISPR Journal**.

T Vanderstichele, KL Burnham, N de Klein, M Tardaguila, B Howell, K Walter, K Kundu, **J Koeppel**, […], E Davenport (2024) Misexpression of inactive genes in whole blood is associated with nearby rare structural variants. **AJHG.**

M Slabicki*, Z Kozicka*, G Petzold*, YD Li, M Manojkumar, RD Bunker, KA Donovan, QL Sievers, **J Koeppel**, D Suchyta, AS Sperling, EC Fink, JA Gasser, LR Wang, SM Corsello, RS Sellar, M Jan, D Gillingham, C Scholl, S Frohling, TR Golub, ES Fischer, NH Thoma and BL Ebert (2020). "The CDK inhibitor CR8 acts as a molecular glue degrader that depletes cyclin K". **Nature**.

A Pallaseni*, EM Peets*, **J Koeppel**, J Weller, T Vanderstichele, UL Ho, L Crepaldi, J Leeuwen, F Allen, L Parts (2022). "Predicting base editing outcomes using position-specific sequence determinants". **Nucleic acids research.**

L Nitsch*, P Jensen*, H Yoon, **J Koeppel**, SSR Burman, ES Fischer, C Scholl, S Fröhling, M Słabicki (2022). "BTB^{BCL6} dimers as building blocks for reversible drug-induced protein oligomerization". **Cell reports methods.**

JC Fowler*, C King, C Bryant, M Hall, R Sood, SH Ong, E Earp, D Fernandez-Antoran, **J Koeppel**, SC Dentro, D Shorthouse, A Durrani, K Fife, E Rytina, D Milne, A Roshan, K Mahububani, K Saeb-Parsy, BA Hall, M Gerstung, PH Jones (2020). "Selection of oncogenic mutant clones in normal human skin varies with body site". **Cancer Discovery.**

RS Sellar*, AS Sperling*, M Słabicki, JA Gasser, ME McConkey, KA Donovan, N Mageed, DN Adams, C Zou, PG Miller, RK Dutta, S Boettcher, AE Lin, B Sandoval, VA Quevedo Barrios, V Kovalcik, **J Koeppel**, EK Henderson, EC Fink, L Yang, A Chan, SP Pokharel, EJ Bergstrom, R Burt, ND Udeshi, SA Carr, ES Fischer, C Chen, BL Ebert (2022) "Degradation of GSPT1 causes TP53-independent cell death in leukemia while sparing normal hematopoietic stem cells". **The Journal of Clinical Investigation**.

J Wu*, S Ma*, R Sandhoff*, Y Ming, A Hotz-Wagenblatt, V Timmerman, N Bonello-Palot, B Schlotter-Weigel, M Auer-Grumbach, P Seeman, WN Loscher, M Reindl, F Weiss, E Mah, N Weisshaar, A Madi, K Mohr, T Schlimbach, RM Velasco Cardenas, **J Koeppel**, F Grunschlager, L Muller, M Baumeister, B Brugger, M Schmitt, G Wabnitz, Y Samstag and G Cui (2019). "Loss of Neurological Disease HSAN-I-Associated Gene SPTLC2 Impairs CD8(+) T Cell Responses to Infection by Inhibiting T Cell Metabolic Fitness". **Immunity**.

Reviews

J Koeppel, J Weller, T Vanderstichele, L Parts (2024). "Engineering structural variants to interrogate human genome function". **Nature Genetics**.

Inventions

L Parts, J Koeppel, T Ellis, R Ferreira, GM Church (2023) "Methods for modifying a cell genome". Provisional US Patent App.

B Ebert, M Slabicki, ES Fischer, H Yoon, J Koeppel (2022) "Chemically inducible polypeptide polymerization". US Patent App. 17/750,061

Oral presentations and posters

Contributed talk, Randomizing the human genome by engineering recombination between repeat elements. Keystone Symposia - Precison genome engineering conference (2024). Banff, Canada.

Invited talk, Engineering genomes with prime editing. MPI for Evolutionary Anthropology – Pääbo/Riesenberg department seminar (2023). Leipzig, Germany

Contributed poster, Randomizing the human genome by engineering recombination between repeat elements. Cold Spring Harbor Laboratories Biology of Genomes conference (2023). Cold Spring Harbor, US.

Invited talk, Prediction of prime editing insertion efficiencies using sequence features and DNA repair determinants. Harvard Medical School – Genome Engineering Seminar Series (2023). Virtual.

Contributed talk, Randomizing gene regulatory regions using prime editing. Wellcome Connecting Science - Synthetic Biology for Future Health (2023). Cambridge, UK.

Contributed poster, Randomizing gene regulatory regions using prime editing. American Institute of Chemical Engineers - International Mammalian Synthetic Biology Workshop (2022). Edinburgh, UK.

Contributed talk, Predicting the efficiency of writing short sequences into the genome using prime editing. Keystone Symposia - Precison genome engineering conference. *Keystone Symposia scholarship awarded* (2022). Keystone, USA.

Contributed poster, Predicting the efficiency of writing short sequences into the genome using prime editing. Keystone Symposia - Precison genome engineering conference (2022). Keystone, USA.

Contributed talk, Determinants of efficiency for writing small sequences into the genome using prime editing. Wellcome Connecting Science - CRISPR and Beyond: Perturbations at Scale to Understand Genomes 2021 (2021). Cambridge, UK.

Invited talk, Determinants of efficiency for writing small sequences into the genome using prime editing. Atlas of Variant Effect Alliance - Variant effect seminar series (2021). Virtual. [LINK]

Invited talk, Chemically induced degradation of the oncogenic transcription factor BCL6. Boehringer Ingelheim webinar (2021). Virtual. [LINK].

Invited talk, Small-molecule-induced protein polymerization. Dana-Farber Cancer Institute - Targeted protein degradation seminar series (2020). Virtual. [LINK]

Contributed poster, Functional dissection of protein degradation. Broad Institute of MIT and Harvard – Cancer Program Poster Session (2019). Boston, USA

Peer review: Helped with reviewing manuscripts at Nature, Nature Biotechnology and Nature Communications